Purdue University ChE 348: Chemical Reaction Engineering (4 credit), Spring 2026 Syllabus

Instructor: Brian Tackett
Assistant Professor
Chemical Engineering
Email: bmtacket@purdue.edu
Office: FRNY 2158
Phone: 765-496-7235
Office Hours: Wednesdays 5:00 – 6:30 pm, FRNY 2142

Teaching Assistants: Wei-Ting Chung (chung380@purdue.edu)
Reagan Huckabee (rhuckabe@purdue.edu)
Jyotisman Rath (rath8@purdue.edu)

TA Office Hours: Wei-Ting:
Reagan:
Jyotisman:

Lecture Hours: Mon/Wed/Fri
10:30 – 11:20 AM
WALC 3087
Synchronous, in-person

Recitation Hours: **All in HAMP 1266**		
	Recitation Section 1:	Tues. 9:30 – 10:45 AM
	Recitation Section 2:	Tues. 11:30 AM – 12:45 PM
	Recitation Section 3:	Tues. 1:30 – 2:45 PM
	Recitation Section 4:	Thur. 9:30 – 10:45 AM
	Recitation Section 5:	Thur. 11:30 AM – 12:45 PM

Laboratory Hours: **FRNY G111** **Feb. 10/12, Feb. 24/26, Mar. 24/26, Apr. 14/16**		
	Lab Section 1:	Tues. 9:30 – 11:20 AM
	Lab Section 2:	Tues. 11:30 AM – 1:20 PM
	Lab Section 3:	Tues. 1:30 – 3:20 PM
	Lab Section 4:	Thur. 9:30 – 11:20 AM
	Lab Section 5:	Thur. 11:30 AM – 1:20 PM

Website: Brightspace (CHE 348). All course materials will be posted on Brightspace. It is your responsibility to keep up-to-date with all material posted online. All class announcements will be posted via Brightspace.

Required Text: H. S. Fogler, Elements of Chemical Reaction Engineering, **Sixth or Seventh Edition**, Prentice Hall, New York. ISE 9780135486498.

E-text is also acceptable and is available for free via the Purdue Library: https://purdue.primo.exlibrisgroup.com/permalink/01PURDUE_PUWL/uc5e95/alma99171049791801081

Textbook companion website:

https://websites.umich.edu/~elements/7e/index.html
Contains extra problems, concept questions, interactive lessons, etc.

Course Description: Application of kinetic rate equations, mass balances and energy balances to the analysis and design of chemical reactors involving homogeneous and heterogeneous chemical reactions. Chemical equilibria, kinetic rate equations for homogeneous and heterogeneously catalyzed reactions, design of ideal isothermal reactors, effects of non-isothermal operation, effects of diffusion in porous catalysts and non-ideal mixing in continuous flow reactors.

Prerequisites: Chemical Engineering 211, Math Selective II, Chemistry 261 (concurrent)

Learning Outcomes: By the end of the course, students will be able to:

1. Apply fundamental material balances to derive the design equations for CSTR, PFR, PBR, and batch reactors.
2. Derive rate expressions from both elementary steps and reaction kinetics data for both homogeneous and heterogeneous reactions.
3. Use material and energy balances with kinetic data for both single and multiple reactions to design and analyze the behavior of isothermal and non-isothermal reactors.
4. Utilize effectiveness factors governing the coupling of reaction and diffusion in the description of heterogeneously catalyzed reactions.
5. Work professionally and ethically in teams to conduct reaction-based laboratory experiments.
6. Effectively report results in written form and practice safety as an integral part of laboratory work.
7. Apply appropriate computational tools for the solution of chemical reaction engineering problems.

Technology Requirement: Students are required to use computational software to solve reaction engineering problems for homework, project, and laboratory sessions. The TAs will provide training and support for **Python** and **Polymath** software. Other programs, such as MATLAB and Mathematica, may also be sufficient for solving the algebraic and ordinary differential equations encountered in this course, but the TAs will not provide training for these methods. So it is recommended to use either Python or Polymath. Both programs are available on any FRNY computer lab machine (FRNY 1014, FRNY 1022, FRNY 1033, FRNY G023). You can also download software to use Python or Polymath on your personal computer:

- Python: is open source, free software. The most convenient way to get started with Python is to download Anaconda software from Anaconda.com. Follow download/install instructions on the website. Once installed, you can select the environment to run Python in. Spyder is a Python environment similar to MATLAB. Jupyter Notebooks is a Python environment in your web browser. Either will work.

- Polymath: is proprietary software developed specifically for the type of problems encountered in this course. You can access this software with your Purdue credentials in AppsAnywhere by following this link: https://appsanywhere.ecn.purdue.edu/

Recitations:

There are five recitation sections held each week. **You may only attend the recitation section for which you are registered**. Recitation will not be held during the weeks of laboratory experiments.

Laboratory:

The Fundamentals Laboratory (FRNY G111) will be used to support this course. There are four weeks throughout the semester during which the lab periods, but not recitation, will be held. The lab period dates and the lab report due dates are listed in the calendar at the end of the syllabus. **These dates are set, and it is not possible to schedule makeup labs. Please make every effort to attend your designated lab time.** If you are physically unable to attend lab, **you must notify Prof. Tackett with a valid reason for your absence prior to the start of your lab session,** otherwise this will be considered an unexcused absence

Lab reports are due two weeks following the lab experiment, by 11:59pm eastern time (on Thursdays). One lab report per group must be uploaded to Gradescope.

The lab manual containing detailed information about the experiments and lab report guidelines can be found at the following link, and accessed using your Purdue career account login credentials: https://engineering.purdue.edu/Intranet/Groups/Schools/ChE/FundamentalsLabMaterials

If you need assistance gaining access to the Fundamentals Lab materials, more information can be found on Brightspace.

It is important that you read the lab manual one week in advance of your experiment.

Part of the recitation sessions one week before the lab will also include a pre-lab activity, to discuss and prepare for the upcoming lab. For each lab report, each team member will be required to fill out a peer evaluation form and submit this via Gradescope, which will be used to adjust the lab grades assigned to each team member.

A few reminders:

- Please read, sign, and bring the Fundamental Laboratory (FL) safety contract with you on the first day in the FL. You are required to complete the contract each time you start a new course in the FL.

- You are now required to bring your own safety glasses to the FL. The safety glasses should have side shields. You can use

the goggles purchased for chemistry labs. If you need to purchase safety glasses, Follett's (across from Mackey Arena on Northwestern) stocks them.

- No coats, backpacks, or large bags are allowed into the FL. Students will not be permitted to leave their coats and backpacks outside the FL door, as was permitted in the past. You should have received an e-mail from Sandy Hendryx with your locker assignment for the year and locker combination. If you did not receive an e-mail with this information, please contact Sandy Hendryx (hendryxs@purdue.edu).

- The Fundamentals Lab is fully scheduled during the semester and therefore, **we cannot conduct makeup lab sessions.** If you have an excused absence from a lab session, you may compose the report with your group members as normal – but you must receive written permission from Prof. Tackett to do so. Please show up 5 minutes early to your lab session. If you show up late to your lab period, you will be dismissed, and there will be no chance to make up missed labs. Missed lab periods without a valid reason will result in a zero grade. If you miss more than two lab periods without a valid reason, then you will be given an I or F grade for the course.

Course Grading: The final course grade will be determined by the following:

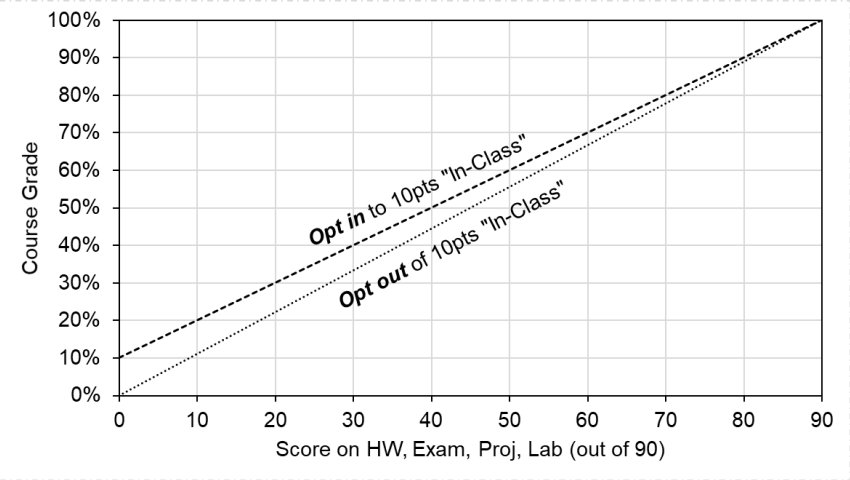
- Lab Reports (4): 20 points
- Exams (3): 35 points
- Homeworks (13): 25 points
- Group Project (1): 10 points
- In-class: (10 points)

Individual assessments for each category are given equal weight (i.e. all 3 exams are weighted the same, all 13 homeworks are weighted the same, etc.). Individual assignments will be graded out of 100%, and the average percent for each category will determine the total class points for that category (e.g. an 80% average on exams yields 26.4/33 points).

10 Point In-class Option: This 10 points is for participating in "in-class" quizzes, clicker questions, and discussion. Credit is earned based on participation in these activities, and you can only earn the full 10 points by participating in >75% in-class activities. If you earn the 10 points, your grade will be calculated as the ***opt in*** scenario below. If you participate in <75% in-class activities, you will not be penalized, and your grade will be calculated as the ***opt out*** scenario below. You will automatically earn the 10 opt in points if you meet the >75% requirement (you do not have to request it). This grading scheme is meant to accommodate various learning styles.

Opt in: course grade = [(x + 10)/100] * 100%

Opt out: course grade = [x/90] * 100%
x = cumulative score on HW, Exams, Project, and Lab Reports



All grades will be available on Brightspace so that you can monitor your progress throughout the semester. Grades for individual homeworks, lab reports, and exams will **not** be adjusted by curving or scaling.

There is no preset distribution of final grades. The grading will reflect demonstrated student capability relative to an absolute performance standard that is expected of all Purdue ChE students, rather than a scale or curve that compares students to a mean performance metric on any evaluation vehicle. In practice, this means that if all students in the class demonstrate a high level of mastery of the course content, then all course grades could be A marks.

If your final numerical grade is greater than or equal to the following percentages, your letter grade is guaranteed to be at least:

- A: ≥90%
- B: ≥80%
- C: ≥70%
- D: ≥60%

Final numerical grades for the entire class may be scaled up (but never down). **Plus and minus modifiers will be used to determine final grades.**

Exams:

Exam 1: Mon. Feb. 16, 8:00pm – 9:15pm, PHYS 112
Exam 2: Mon. Apr. 6, 8:00pm – 9:15pm, PHYS 114
Exam 3/Final: TBD

If students require approved accommodations for exams, these should be scheduled to be taken at the DRC in Stuart Hall. **Students requiring accommodations are responsible for scheduling with the DRC prior to exam dates to ensure DRC availability.**

Homework:

Homework will be assigned via Gradescope, and will be **due on Gradescope at 11:59PM on Thursdays**. Late homework submissions will be assigned a zero score. HW solutions will be posted to Brightspace when homeworks are graded (~1 week after due date).

Each HW problem will be graded on a scale of 2. 1 point will be earned for an honest attempt to solve the problem (beyond rewriting the problem statement). The remaining 1 point will be earned by correctly solving the problem, with partial credit given accordingly.

Scores on all HW assignments will be counted toward the course grade. **No HW grades will be dropped**.

HW 0 counts for 2 full HW grades and consists of course evaluations: mid-term, and final evaluations. Each one must be completed by their respective dates in the course schedule in order to receive full credit for HW 0. These evaluations are critical to the success of the course and should be taken seriously.

Many homework problems will be assigned from the textbook. **Use of the textbook solutions manual is not permitted.**

You may work on homework with other students in class, but you must submit your own individual assignment. **You must also list the names of each person with whom you collaborated on a homework assignment.**

Use of AI Chatbots for HW: Similar to collaborating with other students, you must specify if AI tools were used to help solve homework problems, and you must prepare your own original solution – do not copy and paste chat outputs. Please add an appendix to the end of your assignment with your chat log.

*****All HW problems that require python/polymath or other solvers MUST INCLUDE A PRINTOUT OF ORIGINAL CODE AND OUTPUT IN THE HW SUBMISSION.** Problems submitted with missing code will be assigned 0 points.

***If a problem asks for a plot or graph of a specific scenario, you must provide actual plotted data, not a sketch. Submissions for such problems with absence of plotted data will be assigned 0 points.

Regrade Requests:

You have one week after receiving a graded assignment to submit a regrade request. Requests must be made in-person to Prof. Tackett before or after lecture, or during office hours. Email requests may also be made to schedule an in-person meeting about regrading, as long as it is within one week of receiving the graded item.

Official Purdue University Student Policies

Student Expectations: This is a 4 credit hour course, and it is expected that each student will spend 12 hours each week, including class time, on homework assignments, studying and reading the course textbook.

Student Conduct and Academic Integrity: University policy states that it is the responsibility of all students to attend all class sessions. Each student is expected to come to class on time and not disrupt the class. Each student is expected to follow Purdue's codes of student conduct and behave in a professional manner (https://www.purdue.edu/odos/academic-integrity). The rights of students in violation of the code of conduct are outlined. Each student is expected to exhibit consideration and respect towards the other students, the graders, the teaching assistants (TAs), and the faculty. Each student is expected to exhibit a positive attitude. Your conduct will be a factor in awarding grades to students between two letter grades.

Purdue University's student conduct policy specifically addresses academic dishonesty and integrity (http://www.purdue.edu/odos/osrr/academicintegritybrochure.php). All incidents of academic dishonesty will be reported to the Dean of Students. **Such incidents *include*:**

i) possessing or accessing, in hardcopy or electronic form, the solution manual to the course text or to the exams,
ii) claiming credit for work (either HW or exam work) that is not your own original work, and
iii) enabling another student to create HW or exam work that is not their original work.

Instructors' Commitment: Your instructor will: 1) be courteous, punctual, well-organized, and prepared for lecture and other class activities; 2) answer questions clearly in class or arrange for detailed discussions out of class if in-class answers are not suitably clear; 3) be available during office hours or notify you beforehand if they are unable to keep them; 4) provide a suitable guest lecturer or a recorded video lecture when traveling; and 5) grade uniformly and consistently to the posted guidelines. I strongly encourage you to discuss academic or personal questions with the course instructor during office hours or via email. These discussions need not be limited to ChE 34800 content.

Use of Copyrighted Materials: Among the materials that may be protected by copyright law are the lectures, notes, and other material presented in class or as part of the course. All materials presented by an instructor are protected by copyright unless the instructor has stated otherwise. Students enrolled in, and authorized visitors to, Purdue University courses are permitted to take notes, which they may use for individual/group study or for other non-commercial purposes reasonably arising from enrollment in the course or the University generally.

Notes taken in class are, however, generally considered to be "derivative works" of the instructor's presentations and materials, and they are thus subject to the instructor's copyright in such presentations and materials. No individual is permitted to sell or otherwise barter notes, either to other students or to any commercial concern, for a course without the express written permission of the course instructor.

Accessibility and Accommodations: Purdue strives to make learning experiences as accessible as possible. If you anticipate or experience physical or academic barriers based on

disability, you are welcome to let Prof. Tackett know to discuss options. You are also encouraged to contact the Disability Resource Center at: drc@purdue.edu or by phone: 765-494-1247.

Grief Absence Policy for Students: Purdue recognizes that a time of bereavement is very difficult for a student. Purdue therefore provides the following rights to students facing the loss of a family member through the Grief Absence Policy for Students (GAPS). GAPS Policy: Students will be excused for funeral leave and given the opportunity to earn equivalent credit and to demonstrate evidence of meeting the learning outcomes for missed assignments or assessments in the event of the death of a member of the student's family.

Please visit the University's website for additional information:
http://www.purdue.edu/studentregulations/regulations_procedures/classes.html

Mental Health Statement: If you find yourself beginning to feel some stress, anxiety and/or feeling slightly overwhelmed, try WellTrack, https://purdue.welltrack.com/. Sign in and find information and tools at your fingertips, available to you at any time.

If you need support and information about options and resources, please see the Office of the Dean of Students, http://www.purdue.edu/odos, for drop-in hours (M-F, 8am- 5pm).

If you are struggling and need mental health services, Purdue University is committed to advancing the mental health and well-being of its students. If you or someone you know is feeling overwhelmed, depressed, and/or in need of mental health support, services are available. For help, such individuals should contact Counseling and Psychological Services (CAPS) at (765)494-6995 and http://www.purdue.edu/caps/ during and after hours, on weekends and holidays, or by going to the CAPS office of the second floor of the Purdue University Student Health Center (PUSH) during business hours.

Violent behavior policy: Purdue is committed to providing a safe and secure campus environment for members of the university community. Purdue strives to create an educational environment for students and a work environment for employees that promote educational and career goals. Violent Behavior impedes such goals. Therefore, Violent Behavior is prohibited in or on any University Facility or while participating in any university activity.

Please visit the University's website for additional information:
http://www.purdue.edu/policies/facilities-safety/iva3.html

Nondiscrimination Statement: Purdue is committed to maintaining a community which recognizes and values the inherent worth and dignity of every person; fosters tolerance, sensitivity, understanding, and mutual respect among its members; and encourages each individual to strive to reach their own potential. In pursuit of its goal of academic excellence, the University seeks to develop and nurture diversity. The University believes that diversity among its many members strengthens the institution, stimulates creativity, promotes the exchange of ideas, and enriches campus life.

Purdue views, evaluates, and treats all persons in any University related activity or circumstance in which they may be involved, solely as individuals on the basis of their own personal abilities, qualifications, and other relevant characteristics.

Purdue prohibits discrimination against any member of the University community on the basis of race, religion, color, sex, age, national origin or ancestry, genetic information, marital status,

parental status, sexual orientation, gender identity and expression, disability, or status as a veteran. The University will conduct its programs, services, and activities consistent with applicable federal, state, and local laws, regulations and orders and in conformance with the procedures and limitations as set forth in Purdue's Equal Opportunity, Equal Access and Affirmative Action policy which provides specific contractual rights and remedies. Additionally, the University promotes the full realization of equal employment opportunity for women, minorities, persons with disabilities and veterans through its affirmative action program.

Any question of interpretation regarding this Nondiscrimination Policy Statement shall be referred to the Vice President for Ethics and Compliance for final determination.

Please visit the University's website for additional information:
http://www.purdue.edu/purdue/ea_eou_statement.html

Campus Emergency: In the event of a major campus emergency, course requirements, deadlines and grading percentages are subject to changes that may be necessitated by a revised semester calendar or other circumstances beyond the instructor's control. Relevant changes to this course will be posted onto the course website or can be obtained by contacting the instructors or TAs via email or phone. ***You are expected to read your @purdue.edu email on a frequent basis.***

ChE 348 – Spring 2026 - Tentative Course Schedule

Assignment Due	Date	Topics
		Ch. 1: Definitions and mole balance equations
	Jan. 12	Lecture (Introduction to Course)
	Jan. 13	Recitation (Sections 1, 2, 3)
	Jan. 14	Lecture
	Jan. 15	Recitation (Sections 4, 5)
	Jan. 16	Lecture
		Ch. 2: Conversion and design equations
	Jan. 19	No Class (MLK Jr. Day)
	Jan. 20	Recitation (Sections 1, 2, 3)
	Jan. 21	Lecture
HW 1	Jan. 22	Recitation (Sections 4, 5)
	Jan. 23	Lecture
		Ch. 3: Rate laws
	Jan. 26	Lecture
	Jan. 27	Recitation (Sections 1, 2, 3)
	Jan. 28	Lecture
HW 2	Jan. 29	Recitation (Sections 4, 5)
	Jan. 30	Lecture
		Ch. 7: Experimental evaluation of rate parameters
	Feb. 2	Lecture
	Feb. 3	Recitation / pre-lab (Sections 1, 2, 3)
	Feb. 4	Lecture
HW 3	Feb. 5	Recitation / pre-lab (Sections 4, 5)
	Feb. 6	Recorded Lecture
		Ch. 4: Stoichiometry
	Feb. 9	Lecture
	Feb. 10	Lab #1 (Sections 1, 2, 3)
	Feb. 11	Lecture
HW 4	Feb. 12	Lab #1 (Sections 4, 5)
	Feb. 13	Lecture Cancelled due to Evening Exam 1

Assignment Due	Date	Topics
		Ch. 5: Isothermal reactor design
Exam 1	Feb. 16	Exam 1 (evening) -- Ch. 1,2,3,4,7
	Feb. 17	Recitation / pre-lab (Sections 1, 2, 3)
	Feb. 18	Lecture
	Feb. 19	Recitation / pre-lab (Sections 4, 5)
	Feb. 20	Lecture
		Ch. 6: Semi-batch and membrane reactors
	Feb. 23	Lecture
	Feb. 24	Lab #2 (Sections 1, 2, 3)
	Feb. 25	Lecture Cancelled due to Evening Exam 2
Lab 1 Report/ HW 5	Feb. 26	Lab #2 (Sections 4, 5)
	Feb. 27	Lecture
		Ch. 8: Multiple reactions
	Mar. 2	Lecture
	Mar. 3	Recitation (Sections 1, 2, 3)
	Mar. 4	Lecture
HW 0 / HW 6	Mar. 5	Recitation (Sections 4, 5)
	Mar. 6	Lecture
		Ch. 11: Energy balance and adiabatic reactors
	Mar. 9	Lecture
	Mar. 10	Recitation (Sections 1, 2, 3)
	Mar. 11	Lecture
Lab 2 Report/ HW 7	Mar. 12	Recitation (Sections 4, 5)
	Mar. 13	Lecture
		Spring Break
	Mar. 16	No Class (Spring Break)
	Mar. 17	No Recitation (Spring Break)
	Mar. 18	No Class (Spring Break)
	Mar. 19	No Recitation (Spring Break)
	Mar. 20	No Class (Spring Break)

Assignment Due	Date	Topics
		Ch.11 / Ch. 12: Steady-state reactors with heat exchange
	Mar. 23	Lecture
	Mar. 24	Lab #3 (Sections 1, 2, 3)
	Mar. 25	Recorded Lecture
HW 8	Mar. 26	Lab #3 (Sections 4, 5)
	Mar. 27	Lecture
		Ch. 12 / Ch. 13: Unsteady-state non-isothermal reactors
	Mar. 30	Lecture
	Mar. 31	Recitation (Sections 1, 2, 3)
	Apr. 1	Lecture
HW 9	Apr. 2	Recitation (Sections 4, 5)
	Apr. 3	Lecture
		Ch. 10: Catalysis and catalytic reactors
Exam 2	Apr. 6	Exam 2 (evening) – Ch. 5, 6, 8, 11, 12
	Apr. 7	Recitation (Sections 1, 2, 3)
	Apr. 8	Lecture
Lab 3 report	Apr. 9	Recitation (Sections 4, 5)
	Apr. 10	Lecture
		Ch. 9: Mechanisms and bioreactors
	Apr. 13	Lecture
	Apr. 14	Lab #4 (Sections 1, 2, 3)
	Apr. 15	Lecture
HW 10	Apr. 16	Lab #4 (Sections 4, 5)
	Apr. 17	Lecture
		Ch. 14: Mass transfer in reactors
	Apr. 20	Lecture
	Apr. 21	Recitation (Sections 1, 2, 3)
	Apr. 22	Lecture
HW 11	Apr. 23	Recitation (Sections 4, 5)
	Apr. 24	Lecture

Assignment Due	Date	Topics
		Ch. 15 Diffusion and reaction
	Apr. 27	Lecture
	Apr. 28	Recitation (Sections 1, 2, 3)
	Apr. 29	Lecture
Lab 4 report	Apr. 30	Recitation (Sections 4, 5)
HW 0 / Group Project	May. 1	Lecture

Exam 3/ Final	TBD	Ch 1-15